

Acme Engineering Handbook

Platform Engineering Standards — Edition 2022

1. Architecture Overview

The Acme platform is a microservices architecture deployed on a self-managed Kubernetes cluster running on AWS EC2. Services communicate via REST over HTTP/1.1 for synchronous calls and RabbitMQ for asynchronous messaging. The primary datastore is PostgreSQL 13, with Memcached for caching. Object storage uses S3.

All services must expose a health endpoint at /health and a metrics endpoint at /metrics. Service discovery is handled by Consul. Configuration is managed through environment variables injected at deploy time from a Consul KV store.

2. Deployment Process

All deployments go through Jenkins pipelines. Production deploys require one approval and may occur any time except the last Friday of the month. A change freeze applies during the final two weeks of December. Rollback must be possible within 15 minutes.

Deployment windows are enforced by a custom Jenkins plugin. Emergency deploys outside the window require VP of Engineering approval and a post-incident review.

3. On-Call Rotation

Each team maintains a primary on-call rotation with weekly shifts. PagerDuty is used for paging. Sev-1 incidents require a response within 30 minutes; Sev-2 within 2 hours. Postmortems are mandatory for Sev-1 incidents only.

On-call compensation is a flat \$200 per week stipend. Comp time is not provided. Managers are responsible for ensuring on-call engineers are not overloaded with project work during their rotation week.

4. Code Review Standards

Every pull request requires at least one approving review. PRs touching authentication or payments require two approvals. Maximum PR size is 800 lines of diff. All PRs must pass linting and unit tests before merge. Integration tests run nightly.

Reviewers are expected to respond to PRs within 24 hours. Authors should not merge their own PRs. Draft PRs are permitted and do not require reviews until marked ready.

5. Observability

All services emit structured logs to a self-hosted Elasticsearch cluster. Metrics are scraped by Prometheus every 30 seconds and visualized in Grafana. Distributed tracing is not yet implemented.

Alerting is managed through PagerDuty and Alertmanager. Every alert must have a corresponding runbook. Alerts that fire more than five times per week without action must be tuned or removed.

6. Testing Standards

Unit test coverage must be at least 70 percent per service. Integration tests must cover critical paths. End-to-end tests are run nightly on staging. Load testing is performed quarterly.